

When Nominal False-Positive Rates Fail: Key- and Length-Dependent Null Behavior of Canonical LLM Text Watermarks

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Abstract—Keyed text-watermark detectors are usually described by a nominal false-positive rate that follows from an idealized null: under the KGW green-list scheme, each scored token is green with probability γ independently of the others, so the detector’s z -statistic is standard normal and $z > 2.326$ is a 1% test. We measure how far that null is from real text. Using exact reproductions of the maintained KGW SelfHash and SynthID-Text references, ten frozen detector keys, and 10,000 unwatermarked SmolLM2-135M outputs scored at 128, 256, and 512 tokens, we find that the nominal 1% KGW threshold has an empirical false-positive rate that depends on the key by two orders of magnitude: from 0.27% to 48.4% [47.4, 49.3] at 512 tokens, growing with length for the inflated keys. The mechanism is simple. Each key induces its own null green rate p_k on real text (0.205 to 0.302 against $\gamma = 0.25$), the resulting shift in mean z grows as $\sqrt{T}$ and predicts the observed shift with $r = 0.9999$, and over-dispersion adds the rest. The same keys scored on 5,000 human-written passages reproduce the key ranking (Spearman $\rho = +0.94$ [+0.76, +0.99]), so the bias is a property of the key against ordinary token statistics, not of the generating model, which only amplifies it. Rescoring those same passages under a second vocabulary, with the keys, parameters and text held fixed, does not narrow the spread of p_k , and over 128 keys the ranking does not survive the change ($\rho = +0.01$ [-0.16, +0.18]): the bias belongs to the pair of key and vocabulary, and a rate measured for one tokenizer does not transfer to another. Per-key, per-length empirical calibration on 5,000 outputs brings every held-out cell to 0.16% to 0.96%, at a cost in detection that a development-scale screen prices at a few points. A prospective familywise acceptance rule fitted at its own boundary nevertheless failed 18 of 60 cells, which we report as a design lesson on validation margin. SynthID-Text’s naive-independence reference stays near 1% throughout. All rates are reported with exact 95% intervals.

Index Terms—language-model watermarking, false-positive control, empirical calibration, KGW SelfHash, SynthID-Text, detector keys

I. INTRODUCTION

A keyed text watermark lets a detector with the key test whether a text was produced by a compatible generation process. In the green-list family introduced by Kirchenbauer et al. [17], a secret key and the preceding context partition the vocabulary at each step into a green list of fraction γ and a red list; generation favours green tokens,

and the detector counts how many of the T scored tokens are green. The detector’s operating point is stated through a z -score whose null distribution is derived under one assumption: that in unwatermarked text each scored token is green with probability exactly γ , independently of every other. Under that assumption z is standard normal and $z > 2.326$ is a one-sided 1% test. Papers, toolkits, and deployment discussions routinely quote this nominal rate as if it were a property of the scheme.

This paper measures how far that null is from the behaviour of a real detector on real text, and explains the gap. We fix the implementation first: the canonical KGW SelfHash and SynthID-Text adapters used here reproduce the maintained reference implementations exactly, so nothing below can be attributed to a mismatch. We then score 10,000 unwatermarked SmolLM2-135M-Instruct outputs at three prefix lengths with ten frozen detector keys per scheme, report every rate with an exact 95% interval, and fit nothing.

That the i.i.d. assumption fails on real text is not new, and it is important to be precise about what was already established. Fernandez et al. [12] demonstrated the failure at scale, over ten master keys and 100,000 Wikipedia sequences of 256 tokens at the same $\gamma = 0.25$, and attributed it to repeated hashing contexts; their remedy is a non-asymptotic test together with a rectified scoring rule that skips repeated contexts. Cai et al. [6] give the general theory of that dependence. Both accounts locate the problem in dependence between scored positions.

Two things follow that have not been shown. First, a second and separate effect survives once dependence is handled: each key colours a different fraction of ordinary text green, so the null mean is displaced rather than merely over-dispersed. This is not a fine distinction. The canonical adapter measured here already applies the rectified scoring of Fernandez et al. [12], skipping repeated contexts, and the nominal 1% threshold still yields a 48.4% false-positive rate on the worst of ten keys. Second, the earlier measurements pool their keys. A spread from 0.27% to 48.4% is invisible in an aggregate over ten keys, and the aggregate is the one quantity a deployment cannot use, because a deployment holds a single key. What is new here is that the failure is a

mixture over keys rather than a uniform bias, that its size and sign are fixed by the key together with the tokenizer’s vocabulary, that it grows with length instead of averaging out, and what correcting it costs in detection. Those are this paper’s contributions, in order of evidential strength.

- 1) **The nominal 1% KGW threshold is key- and length-dependent.** Across ten keys at 512 tokens the empirical false-positive rate spans 0.27% to 48.4% [47.4, 49.3], $n = 10,000$ per cell; 20 of 30 key-by-length cells exceed 1% and the inflated keys get worse with length (Section IV-B). Which key inflates is not portable: a Qwen2.5 replication fails different keys.
- 2) **The mechanism is a key-specific null green rate.** Each key colours a different fraction p_k of real text green, 0.205 to 0.302 against the assumed 0.25. Substituting p_k for γ in the detector’s own formula predicts the mean null z of all 30 cells with $r = 0.9999$, and the shift grows as $\sqrt{T}$. Over-dispersion adds a second, smaller effect. Repetition is not the driver (Section IV-C).
- 3) **The bias belongs to the key and the vocabulary, not the model.** The same keys on 5,000 human-written passages give the same ordering of p_k (Spearman $\rho = +0.94$) and the same worst key, with smaller inflation (Section IV-D). Rescoring those same passages under a three-times-larger vocabulary does not narrow the spread of p_k , and over 128 keys the ranking does not survive ($\rho = +0.01$ [-0.16, +0.18], against +0.94 [+0.76, +0.99] within one vocabulary), with the model, keys, parameters and text all held fixed. The model amplifies the bias but does not choose which key carries it (Section IV-E).
- 4) **Per-key, per-length empirical calibration works, and it is affordable.** Thresholds fitted once on 5,000 outputs and evaluated once on 5,000 disjoint outputs give held-out rates of 0.16% to 0.96% in all 60 scheme-by-key-by-length cells (Section IV-G). A development-scale screen of 50 watermarked outputs per cell prices the cost at a median two points of detection at 512 tokens, gaining detection for the three keys whose threshold it lowers; that screen sizes the remedy and is not a detection claim (Section IV-H).
- 5) **Fitting at an acceptance boundary leaves no familywise margin.** A prospective rule that required all 60 calibrated cells to pass an exact Bonferroni-controlled bound failed 18 cells, although no failed cell’s point estimate exceeded 0.96%. Twenty-eight of 30 KGW thresholds had been fitted at the maximum passing count, where a fresh cell passes with probability about 55% (Section IV-I).

The contribution is deliberately bounded. We do not claim that every implementation called KGW or SynthID behaves alike, that the numbers transfer to other models or languages, or anything about robustness to editing or paraphrase. The detection rates we do report are development-

scale and are there to price the remedy, not to claim a detection result. What we claim is narrower and, we think, more useful: for the named primitives, the nominal false-positive rate is not a property of the scheme, and the evaluation that replaces it must be indexed by key and by length.

II. RELATED WORK

The area is surveyed by Liu et al. [24]; we situate this paper against the strands it touches.

a) *Keyed green-list watermarks.*: Kirchenbauer et al. [17] introduced the green-list scheme and its z-test detector. The follow-up reliability study formalized context-dependent hashing, the SelfHash variant, repeated-context handling, and the dependence of detection on observed length [18]. Both derive the detector’s null from the i.i.d. Bernoulli(γ) assumption examined here. Zhao et al. [36] fix the green list independently of context (Unigram) and prove robustness guarantees under the same style of null. Fernandez et al. [12] observed that the assumption fails on real text and proposed corrected statistics. Their evidence is large and already human-written: ten master keys against 100,000 Wikipedia sequences of 256 tokens at the same $\gamma = 0.25$, with the inflation attributed to repeated hashing contexts and remedied by a non-asymptotic test together with a scoring rule that skips them. Cai et al. [6] give the general theory of that collision-induced dependence. Our measurements begin where those leave off, in three ways. We apply their rectified scoring rather than comparing against its absence, and the inflation persists. We decompose by key instead of pooling, which is what turns a known aggregate failure into a spread of two orders of magnitude across the keys a deployment might actually hold. And the human-text result here is not the first null measured on human text, which theirs already was, but the first *paired* one: the same keys on human and model text, asking whether the ranking of p_k is preserved (Section IV-D).

b) *Distortion-free and cryptographic constructions.*: Kuditipudi et al. [20], Christ et al. [7] and Hu et al. [15] construct watermarks whose output distribution matches the unwatermarked model and whose detectors have exact or provable null behaviour by design. SynthID-Text [9] uses context-dependent seeding and tournament sampling; its production evaluation covers quality, detection, edits, and paraphrasing. These families do not share KGW’s z-test null, which is why our SynthID measurements are a reference rather than a failure and why the key effect is a KGW-specific finding.

c) *Benchmarks and toolkits.*: MarkMyWords [31], WaterBench [34], WaterPark [22], and WaterJudge [25] compare schemes on quality, detection, and tamper resistance; MarkLLM [27] packages many of them behind one interface. These evaluations typically report detection at a scheme’s nominal threshold with one key. The present results show why that convention can misstate the operating

point by an order of magnitude and why the key schedule belongs in the reported condition.

d) *False-positive control.*: Li et al. [21] develop pivotal statistics with explicit type-I error control for watermark detection, and forensic-readiness work argues that error rates must be established before watermark evidence is used in consequential settings [33]. The operating points assumed elsewhere are far stricter than the nominal 1%: attack studies calibrate KGW detectors, the SelfHash variant included, to false-positive rates of 10^{-3} and 10^{-6} on the argument that a false accusation is expensive [16], and evasion studies report detection at a fixed 1% false-positive rate [19]. Both conventions presuppose that the nominal rate is the delivered rate. Fu and Russell [13] show a related but distinct failure, in which the false-positive rate grows with the size of the key space because many keys are tested at once; ours is present with a single key. Xie et al. [35] reach the same operational conclusion we do from a different direction, replacing the nominal threshold with conformal calibration for classroom AI-usage detection. On the negative side of the ledger, Sadasivan et al. [32] and Pang et al. [28] argue that detection faces intrinsic limits under an adversary; our concern is the non-adversarial case, where no attacker is present and the text is simply not watermarked. We supply the empirical side: how large the error is for the most widely used detector, where it comes from, and what calibration removes it.

e) *Tokenization as a source of disparity.*: That a vocabulary partition can create systematic, unequal treatment is established outside watermarking: tokenizers encode the same content at very different lengths across languages, with consequences for cost and context budget [1, 29], and tokenizer choice measurably changes downstream model behaviour [2]. Concurrent work audits watermarking across languages and likewise concludes that detection thresholds must be calibrated per deployment context rather than inherited [26]. Neither holds the key, the parameters and the text fixed while changing only the tokenizer, which is what Section IV-E does to separate the vocabulary from the model.

III. METHODS

A. Implementation Identity

Before any null measurement, the validation suite pinned maintained reference revisions for KGW and SynthID-Text, checked keyed token assignments, reproduced aggregate detector scores from per-position traces, and compared generation step by step. The canonical adapters used throughout this paper are `kgw_author_selfhash_v1` (KGW SelfHash: $\gamma = 0.25$, $\delta = 2.0$, context width 4, repeated contexts ignored) and `synthid_deepmind_hash_v1`. The stock Transformers implementations were retained as separately named secondary variants and are never pooled with canonical scores. The remaining SynthID top- k boundary-tie difference was recorded as a decoder-policy difference.

B. Null Corpus and Detector Statistics

The primary null corpus is 10,000 unwatermarked continuations generated by the pinned SmolLM2-135M-Instruct checkpoint (temperature 0.8, top- k 40, 512 new tokens) from 10,000 distinct Databricks Dolly prompts balanced by task category. The corpus was produced as two disjoint 5,000-prompt splits for the calibration study of Section III-E; because nothing in the nominal-threshold analysis is fitted, both splits are pooled there. Every output was scored at its paired 128-, 256-, and 512-token prefixes by ten frozen detector keys per scheme. Keys within a run therefore score the same texts and are paired, not independent.

For KGW, the detector statistic at a prefix is

$$z = \frac{g - \gamma T}{\sqrt{T\gamma(1 - \gamma)}}, \quad (1)$$

where g is the number of green tokens among the T eligible scored positions (SelfHash skips positions whose context has already been scored, so T is below the prefix length). The nominal one-sided 1% threshold is $z > 2.326$, which is exact only if green membership is i.i.d. Bernoulli(γ) under the null. SynthID-Text’s mean- g detector has no nominal z ; as a reference we report the naive-independence normal approximation $z = (\bar{g} - 0.5)\sqrt{4dT}$ with d watermarking depth, at the same 2.326 cutoff, labelled as such.

All rates are reported with exact two-sided 95% Clopper-Pearson intervals [8]. The exact interval is preferred to the Wald interval throughout because the cells of interest are small- p and, at the smallest cells here, the Wald coverage is erratic in exactly the regime we report [5]. No threshold in Sections IV-B–IV-D was chosen from the data.

C. Secondary Null Runs

Two earlier runs with the same ten keys are reported alongside the primary corpus: a 500-output SmolLM2 development pilot, and an independent balanced 104-output run with Qwen2.5-0.5B-Instruct in a clean environment. Both are development diagnostics; their role here is to show that the primary result reproduces and that which key inflates does not carry from one model and tokenizer to another. That run varies the model and its vocabulary together; the fourth mechanism analysis of Section III-D separates them. A fifth detector-only run widens the key sample rather than the corpus: 128 keys from the same derivation, of which conditions 0–9 are the ten used throughout, scored on 1,500 of the human passages to give the marginal distribution of p_k (Section IV-F).

D. Mechanism Analyses

Four detector-only analyses of the stored token ids ask why the nominal threshold fails. First, for each key k and length we pool green counts over eligible positions to estimate the null green rate p_k and compare the observed mean z with the value Eq. (1) implies when γ is replaced

by p_k . Second, we relate each output’s z to its repeated-4-gram fraction to test whether low-entropy repetition drives the inflation. Third, we score 5,000 de-duplicated human-written Dolly passages (`response` and `context` fields, at least 128 tokens, selected by a hashed seed) with the same ten keys at every prefix length each passage supports. Fourth, we rescore those same passages under a second vocabulary, Qwen2.5-0.5B-Instruct at about 152k tokens against SmolLM2-135M at about 49k, holding the ten hashing keys, the watermark parameters and the text fixed, so that the tokenizer is the only thing that varies. Eligibility at a given prefix length is itself tokenizer-dependent, so each length is restricted to the passages both vocabularies support. No text is generated and no model weights are loaded in any of these analyses.

E. Empirical Calibration and One-Shot Confirmation

The remedy we test is per-key, per-length empirical calibration. Thresholds were fitted once on the 5,000-output calibration split for 60 cells (two schemes, ten keys, three lengths) and then evaluated once on the disjoint 5,000-output confirmation split. Comparisons were strictly $s > t$. KGW thresholds were indexed by key and length; SynthID used one threshold per length equal to the maximum of the ten key-specific calibration candidates, with every key still tested separately at confirmation. The prospective acceptance rule was familywise: a cell passed if its exact one-sided Clopper–Pearson upper bound at a Bonferroni-corrected $\alpha = 0.05/60$ [11] was at most 1%, which allows at most 28 exceedances per cell, and the family passed only if all 60 cells passed. Thresholds were frozen and hashed before the confirmation split was generated, and no threshold was revised afterwards.

IV. RESULTS

A. Canonical Parity Passed

The canonical adapters reproduced both pinned maintained references exactly on the tested detection fixtures and step-by-step generation checks, and aggregate scores reconstructed exactly from exported per-position rows. Every result below is therefore a property of the named primitives rather than of an implementation mismatch.

B. The Nominal 1% KGW Threshold Is Key- and Length-Conditional

Table I and Figure 1 give the empirical false-positive rate of canonical KGW SelfHash at $z > 2.326$ for all 30 key-by-length cells, $n = 10,000$ each. Point estimates exceed 1% in 20 of 30 cells; the exact 95% interval lies entirely above 1% in 19 cells and entirely below 1% in 8. The spread across keys is two orders of magnitude at every length. Key 07 gives 19.1%, 33.0%, and 48.4% [47.4, 49.3] at 128, 256, and 512 tokens; key 03 gives 0.32%, 0.26%, and 0.27%. The median across keys rises from 3.5% at 128 tokens to 6.9% at 512. Inflated keys therefore get worse with length while conservative keys stay conservative, so no single length

policy repairs the threshold. The 500-output pilot value of 47.0% for key 07 at 512 tokens reproduces as 48.4% at $n = 10,000$.

Which key inflates does not carry across runs (Figure 2). On Qwen2.5-0.5B the worst keys are 04 and 05 at 41.3% each, while key 07, the worst key on SmolLM2, gives 5.8%. The ten keys are the same bit strings in both runs. That run changes the model and its tokenizer together and so cannot say which of the two decides; Section IV-E separates them on identical text and finds the vocabulary sufficient.

SynthID-Text’s naive-independence reference behaves differently: per-cell rates range from 0.74% to 1.58%, 11 of 30 cells sit above 1% by point estimate, and seven have intervals entirely above 1%. This is mild over-dispersion with no key-specific bias, and we report it as a reference rather than as a SynthID failure.

C. Mechanism: Each Key Has Its Own Null Green Rate

The per-key null z distributions at 512 tokens (Figure 3) are approximately normal but shifted and widened relative to the assumed $N(0, 1)$: mean z runs from -1.96 (key 03) to $+2.24$ (key 07), and the standard deviation from 1.26 to 1.58.

The shift has a simple source (Figure 4). Each key induces its own null green-token rate p_k on this model’s text, stable across lengths and ranging from 0.205 to 0.302 at 512 tokens against the assumed $\gamma = 0.25$. Substituting p_k for γ in Eq. (1) predicts a mean null z of $(p_k - \gamma)\sqrt{T}/\sqrt{\gamma(1 - \gamma)}$, which grows as $\sqrt{T}$. The observed mean z across all 30 cells matches this prediction with $r = 0.9999$. A second, smaller effect is over-dispersion: the standard deviation of z rises from 1.07–1.27 at 128 tokens to 1.26–1.58 at 512, which is why key 06 ($p_k = 0.249$) still shows 3.7% at 512 tokens. A normal with each cell’s observed mean and standard deviation reproduces every cell’s rate within 1.5 percentage points.

The failure is thus the standard z -test’s i.i.d. Bernoulli(γ) null being wrong for real text, with the sign and size of the error set by which frequent (context, token) pairs each key colours green. This is the phenomenon Fernandez et al. [12] documented for KGW-style detectors; here it is quantified per key and per length on one pinned implementation.

Repetition is not the driver (Figure 5). The outputs are repetitive (median repeated-4-gram fraction 0.23), but for the worst key the correlation between z and repetition is negative (Spearman $\rho = -0.29$): key 07 gives 63% in the least repetitive quartile and 32% in the most repetitive. SelfHash already skips repeated contexts, which is why eligible positions average 363 of 512.

D. Human-Written Text Reproduces the Key Ranking

If p_k were a property of the generating model, human-written text under the same keys would show a different ranking. It does not (Table II, Figure 6). On 5,000 human-written Dolly passages, p_k ranges from 0.216 to 0.280 at 512 tokens and tracks the model’s p_k key for key (Spearman

TABLE I

EMPIRICAL FALSE-POSITIVE RATE (%) OF CANONICAL KGW SELFHASH AT THE NOMINAL ONE-SIDED 1% THRESHOLD ($z > 2.326$) ON 10,000 UNWATERMARKED SMOLLM2-135M-INSTRUCT OUTPUTS PER CELL, WITH EXACT TWO-SIDED 95% CLOPPER-PEARSON INTERVALS. NOTHING IS FITTED.

Key	128 tokens	256 tokens	512 tokens
kgw-00	6.23 [5.76, 6.72]	8.87 [8.32, 9.44]	12.49 [11.85, 13.15]
kgw-01	0.75 [0.59, 0.94]	0.70 [0.55, 0.88]	0.80 [0.63, 0.99]
kgw-02	0.99 [0.81, 1.20]	1.19 [0.99, 1.42]	1.28 [1.07, 1.52]
kgw-03	0.32 [0.22, 0.45]	0.26 [0.17, 0.38]	0.27 [0.18, 0.39]
kgw-04	0.66 [0.51, 0.84]	0.80 [0.63, 0.99]	0.88 [0.71, 1.08]
kgw-05	4.80 [4.39, 5.24]	7.17 [6.67, 7.69]	10.19 [9.60, 10.80]
kgw-06	2.18 [1.90, 2.49]	2.78 [2.47, 3.12]	3.69 [3.33, 4.08]
kgw-07	19.08 [18.31, 19.86]	33.02 [32.10, 33.95]	48.36 [47.38, 49.34]
kgw-08	7.28 [6.78, 7.81]	9.65 [9.08, 10.25]	12.74 [12.09, 13.41]
kgw-09	6.44 [5.97, 6.94]	9.30 [8.74, 9.89]	13.66 [12.99, 14.35]
Median	3.49	4.98	6.94

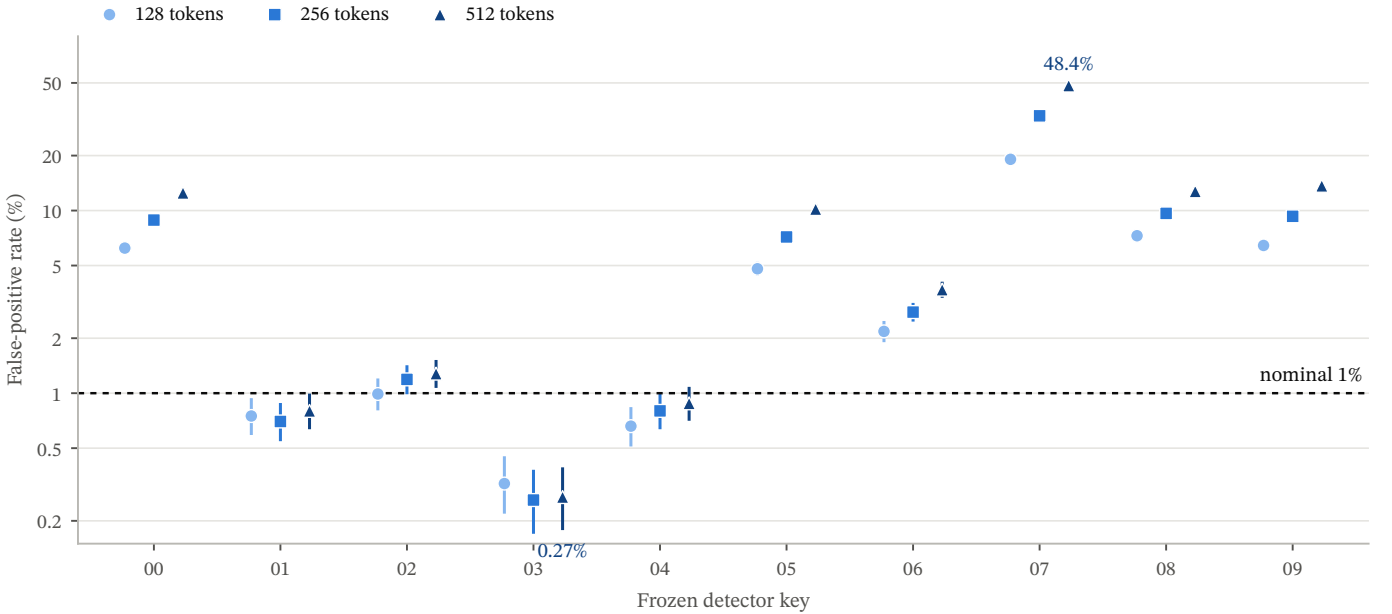


Fig. 1. Canonical KGW SelfHash false-positive rate by frozen detector key and prefix length at the nominal $z > 2.326$ threshold, $n = 10,000$ unwatermarked SmolLM2 outputs per cell, exact 95% intervals, log scale. Prefix length is shaded light to dark. Keys are categorical and are paired within texts, so the points are not independent across keys.

$\rho = +0.94$, $p = 5 \times 10^{-5}$; Pearson $r = +0.96$ over ten keys). Key 07 is the worst key in both populations and keys 03 and 04 the best in both. Human-text rates at the nominal threshold are inflated in the same direction but by less: key 07 gives 9.4%, 16.4%, and 25.5% at 128, 256, and 512 tokens on human text against 19.1%, 33.0%, and 48.4% on model output, and 19 of 30 human cells lie above 1% by point estimate. Human text is also far less repetitive (median repeated-4-gram fraction 0.016), and the inflation persists.

The key-specific bias is therefore a property of each key’s green list against ordinary English token statistics, not of SmolLM2’s output distribution. The model amplifies it, presumably because its low-entropy output concentrates mass

on the same frequent (context, token) pairs. A practical corollary is that a deployment cannot repair the nominal threshold by calibrating on model text alone: human text under the same key inflates too.

E. The Ranking Is Specific to the Vocabulary

The Qwen2.5 replication of Section IV-B changed the model and its tokenizer together, so it cannot say which of the two decides that different keys fail. A natural objection is also that a larger vocabulary should average the key effect away. We separate the two by scoring the identical human passages of Section IV-D under a second vocabulary, Qwen2.5-0.5B-Instruct at about 152k tokens against SmolLM2-135M at about 49k, with the same ten keys, the

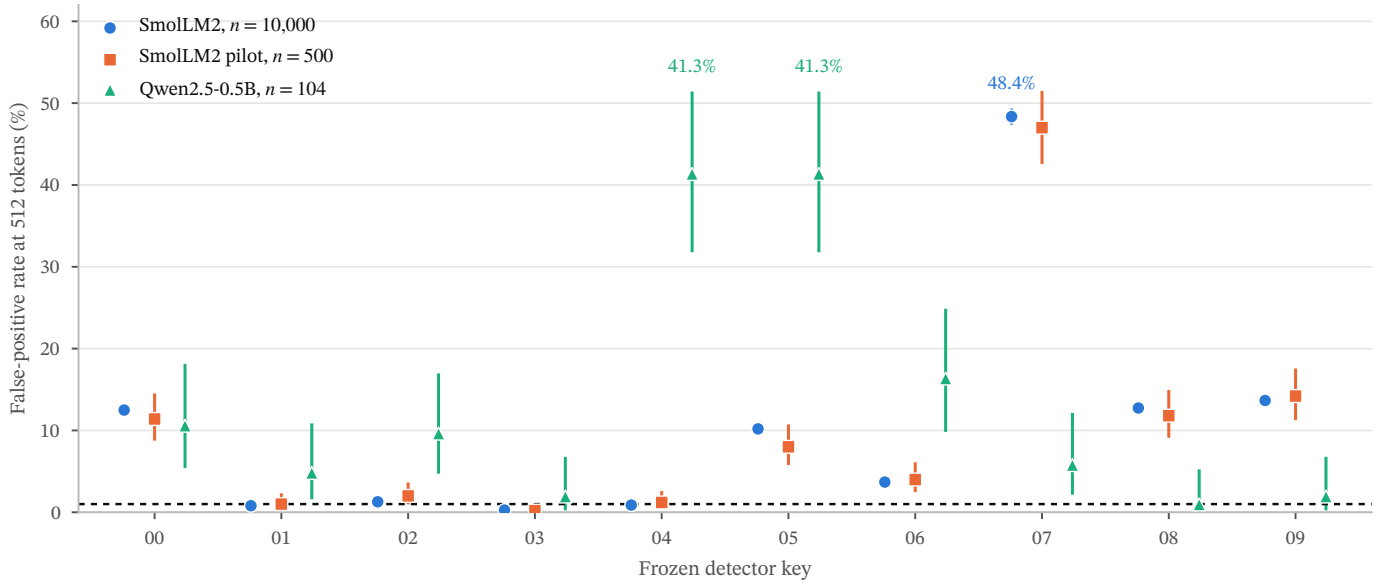


Fig. 2. The same ten keys at 512 tokens in three null runs: the primary SmolLM2 corpus ($n = 10,000$), the SmolLM2 pilot ($n = 500$), and the Qwen2.5-0.5B replication ($n = 104$). The two SmolLM2 runs agree; the Qwen run inflates different keys.

TABLE II
EMPIRICAL FALSE-POSITIVE RATE (%) AT THE NOMINAL $z > 2.326$ THRESHOLD ON HUMAN-WRITTEN DOLLY PASSAGES (HUMAN) AND ON UNWATERMARKED SMOLLM2 OUTPUT (MODEL) FOR THE SAME TEN FROZEN KGW KEYS. EACH HUMAN PASSAGE IS SCORED AT EVERY PREFIX LENGTH IT SUPPORTS, SO n FALLS WITH LENGTH. THE LAST ROWS GIVE THE RANGE OF THE NULL GREEN RATE p_k ACROSS KEYS AND THE CELL SIZES.

Key	128 tokens		256 tokens		512 tokens	
	Human	Model	Human	Model	Human	Model
kgw-00	2.6	6.2	2.7	8.9	2.7	12.5
kgw-01	0.7	0.8	0.5	0.7	0.5	0.8
kgw-02	0.9	1.0	0.9	1.2	1.4	1.3
kgw-03	0.5	0.3	0.9	0.3	1.3	0.3
kgw-04	0.2	0.7	0.2	0.8	0.7	0.9
kgw-05	3.6	4.8	5.4	7.2	7.2	10.2
kgw-06	1.4	2.2	1.0	2.8	1.8	3.7
kgw-07	9.4	19.1	16.4	33.0	25.5	48.4
kgw-08	6.2	7.3	9.2	9.7	14.6	12.7
kgw-09	2.9	6.4	4.8	9.3	6.9	13.7
p_k range	0.217–0.282	0.206–0.301	0.217–0.282	0.206–0.302	0.216–0.280	0.205–0.302
n	5,000	10,000	1,951	10,000	554	10,000

same watermark parameters and the same text. Only the tokenizer changes. Each length is restricted to the passages both vocabularies support, giving $n = 4,857, 1,875$ and 511 at 128, 256 and 512 tokens.

Neither the objection nor the model explanation survives (Figure 7). The spread of p_k across keys does not shrink under the three-times-larger vocabulary; over these ten keys it is 0.065 under SmolLM2 against 0.077 to 0.082 under Qwen2.5, at every length. Nor is there evidence that the ranking survives: over ten keys at 512 tokens the Spearman correlation of p_k between the two vocabularies is $\rho = -0.15$ ($p = 0.68$), against $\rho = +0.94$ between human and model text under one vocabulary. Ten keys is a small

sample for a rank correlation, and we state the limit rather than read a null out of a non-significant test: the 95% interval on ρ here is $[-0.71, +0.53]$, against $[+0.76, +0.99]$ for the human-versus-model comparison. What the data support is that the transfer is not of the strength seen within one vocabulary, not that it is exactly zero. Key 07, the worst key in both populations of Section IV-D, falls to sixth under the Qwen2.5 vocabulary and its rate at the nominal threshold drops from 25.8% to 3.5%; key 05 rises from 6.5% to 67.7% and key 04 from 0.59% to 31.3%. Of 30 key-by-length cells, 19 exceed 1% under the SmolLM2 vocabulary and 24 under the Qwen2.5 vocabulary, so the inflation itself is not repaired by the larger vocabulary; it

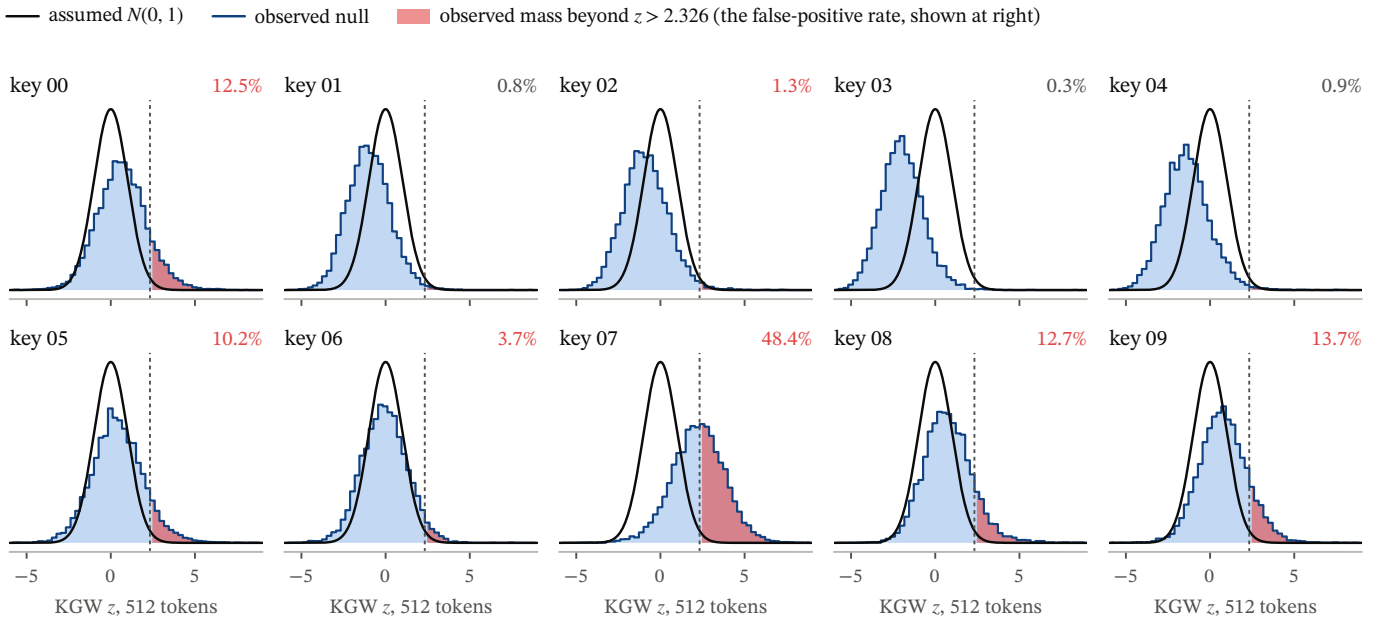


Fig. 3. Null distribution of the KGW z statistic at 512 tokens for each of the ten keys, $n = 10,000$, against the assumed standard normal (solid) and the nominal cutoff (dashed). The shaded area past the cutoff is the mass this key puts beyond the nominal 1% threshold; the percentage at each panel’s top right is that cell’s false-positive rate. Under the assumed null every panel would show the same 1%.

is only redistributed over different keys.

The key effect is therefore a property of the pair (key, vocabulary), not of the key alone and not of the generating model. A green list is a partition of one particular vocabulary, and the fraction of ordinary English it colours green depends on both halves of that pair. This also sharpens the reading of the Qwen2.5 replication: the different failing keys there need not indicate a different frequent-pair distribution in a different model, because changing the tokenizer alone is enough to reshuffle the ranking on text that does not change at all.

Ten keys cannot settle a rank correlation, so we repeated the comparison over the 128 keys of the sweep described below. Both vocabularies score an identical set of passages at an identical set of prefix lengths: the run filters candidates on joint eligibility under both tokenizers, without which the two legs select different passages and the comparison is not paired. This widens the key sample for the paired comparison itself, not only for the marginal distribution of p_k .

The wider sample narrows the interval enough to make the null informative (Figure 8). The correlation is indistinguishable from zero at every length, and the interval is now narrow enough to exclude even a moderate one: $\rho = +0.01$ $[-0.16, +0.18]$ at 128 tokens over 800 passages, $+0.01$ $[-0.16, +0.19]$ at 256 over 313, and -0.02 $[-0.19, +0.15]$ at 512 over 93. The half-width of the interval falls from 0.63 at ten keys to 0.17 at 128, a factor of 3.6. Ten keys admitted everything from a strong negative to a strong positive correlation; 128 keys rule out any transfer stronger than weak. The ranking of keys by p_k does not carry across a

change of vocabulary. The spread itself is unchanged rather than widened: over 128 keys it is 0.098 under SmolLM2 against 0.100 under Qwen2.5 at 128 tokens, and the two are within 0.01 of each other at all three lengths, so the ten-key impression that the larger vocabulary widens the spread does not survive the larger sample. What survives is the claim that matters for the objection: the larger vocabulary does not average the key effect away.

F. How Much of the Key Space Is Affected

Ten keys establish that the effect exists and that its ranking is stable, but not how much of the key space carries it. We widened the sample with the same derivation, so conditions 0–9 reproduce the ten used throughout the paper and appear inside the distribution rather than beside it. The sweep scores 128 keys against 1,500 human-written passages from the corpus of Section IV-D, detector-only and KGW alone.

The spread is not a property of the ten (Figure 9). Across 128 keys p_k runs from 0.217 to 0.315 at 128 tokens and from 0.216 to 0.319 at 512, against the assumed 0.25, with a median of 0.248 and 44.5% of keys above γ . The ten keys of Section IV-B sit inside that distribution and, if anything, slightly below its centre: their median p_k at 128 tokens is 0.250 against 0.248 for all 128. Nothing about them was extreme.

At the nominal threshold the consequence is a majority effect rather than a tail one. At 128 tokens, where every key is scored on all 1,500 passages, the empirical false-positive rate runs from 0.13% to 30.0% with a median of 1.60%; 88 of 128 keys (69%) sit above the nominal 1% by point

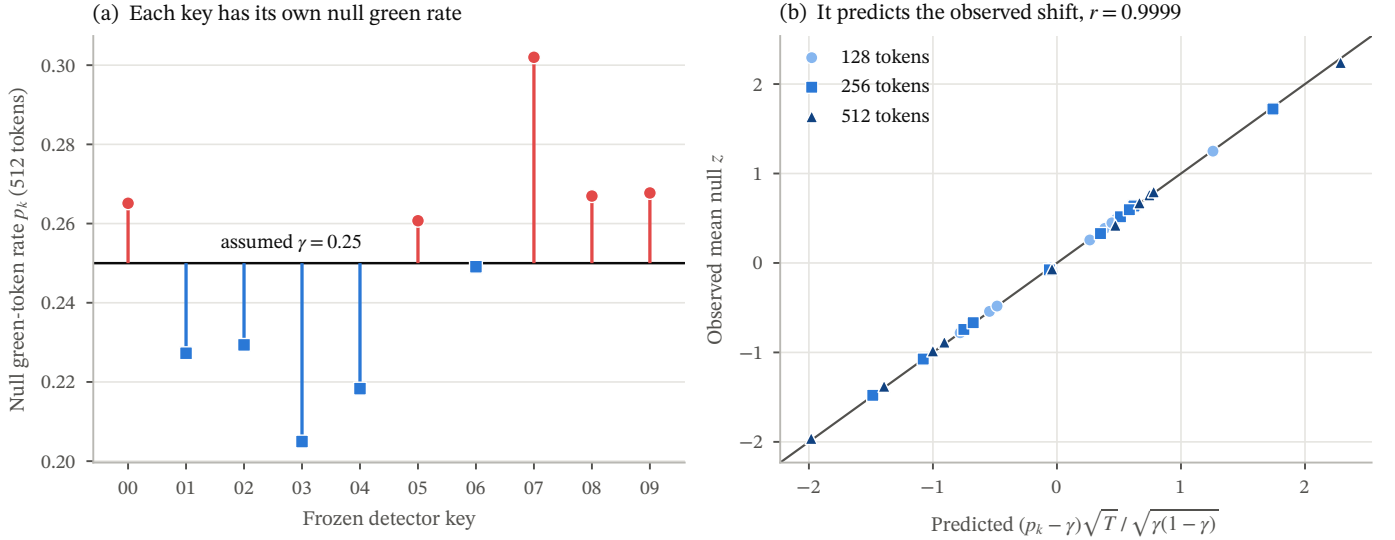


Fig. 4. (a) Null green-token rate p_k for each key at 512 tokens, drawn from the assumed $\gamma = 0.25$; colour marks which side of γ the key falls on and the vertical bars are exact 95% intervals. (b) Observed mean null z for all 30 key-by-length cells against the value predicted from p_k alone, with the identity line.

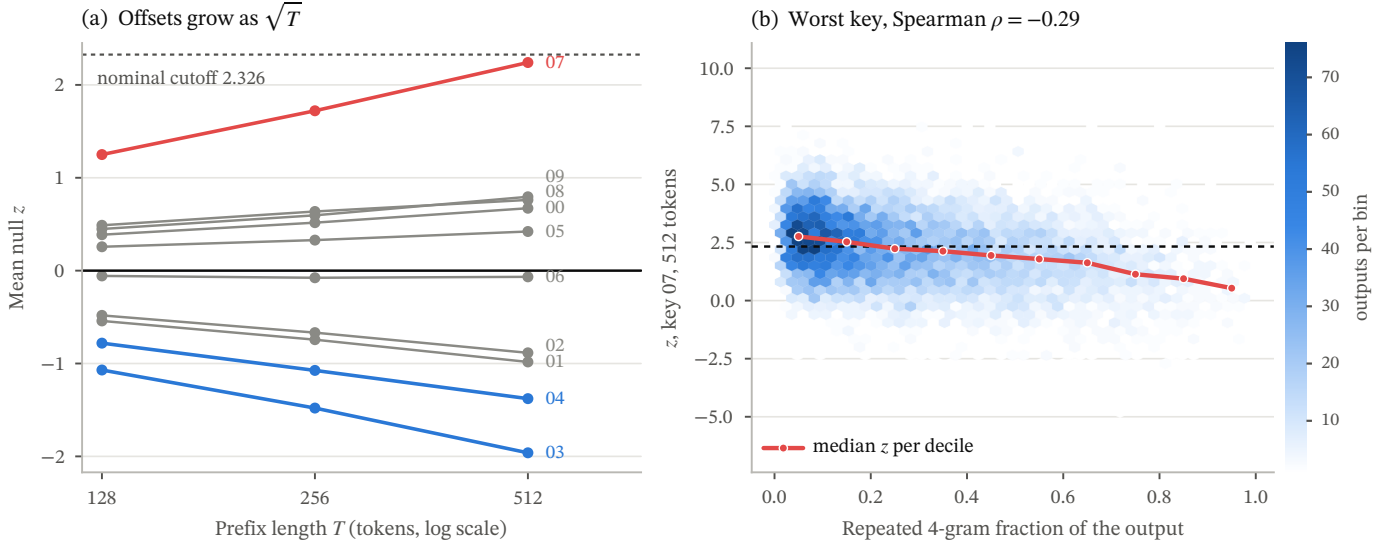


Fig. 5. (a) Mean null z per key against prefix length on a log scale; the offsets grow as $\sqrt{T}$, away from zero in both directions, and only the keys ending beyond $|z| = 1$ are coloured. (b) Density of per-output z for key 07 at 512 tokens against the output’s repeated-4-gram fraction, $n = 10,000$, with the median per decile; more repetition goes with lower z , not higher.

estimate, and 68 (53%) have an exact 95% lower bound that also clears it. At 512 tokens, where 179 passages per key reach that length, the range is 0% to 82.1% with a median of 2.23%, 91 keys (71%) above 1% by point estimate and 57 (45%) by lower bound. The per-key intervals are wider at 512 because the cell is smaller, which is why the conservative count falls while the point estimate rises. The worst key in the sweep is worse than the worst of the original ten: 82.1% against 48.4%.

The practical reading is that a deployment drawing a key at random is more likely than not to operate above the false-positive rate its detector advertises, and that roughly

half of all keys can be shown to do so from 1,500 passages alone. This does not weaken the case for measuring p_k ; it removes the possibility that the ten keys of Section IV-B were an unlucky draw.

G. Per-Key, Per-Length Calibration Brings Held-Out Rates Below 1%

Per-key empirical thresholds are not a new device: Aremu et al. [4] fit them from unwatermarked nulls, one per key, as a component of a defence against watermark forgery. What has not been shown is that they are *necessary* for a single key at the nominal operating point, which

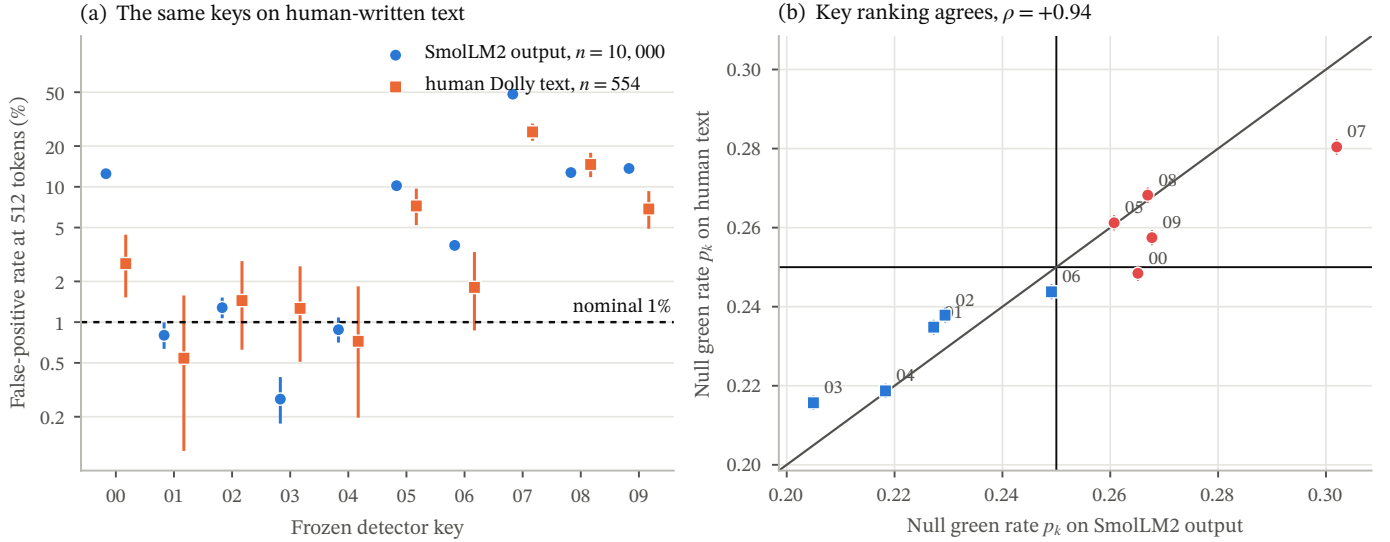


Fig. 6. (a) False-positive rate at the nominal threshold and 512 tokens for the same ten keys on SmolLM2 output ($n = 10,000$) and on human-written Dolly passages ($n = 554$), exact 95% intervals, log scale. The human intervals are wide because the 512-token human cell is small. (b) Per-key null green rate on human text against the same quantity on model output, with exact intervals on both axes; the diagonal is equality and the rules mark $\gamma = 0.25$. Colour is the same side-of- γ encoding as Figure 4a.

is what the spread in Section IV-B implies and what the held-out rates below confirm. The distinction matters for a second reason: that work assumes non-target keys give approximately zero-mean scores under the null, an assumption about cross-key scoring rather than about same-key scoring of unwatermarked text, and our measurements bear only on the latter.

Appendix Table III and Table IV list all 60 calibrated cells. Fitted KGW thresholds at 512 tokens range from $z = 1.49$ (key 03) to $z = 6.40$ (key 07) against the nominal 2.326. On the disjoint confirmation split the held-out false-positive rate of the once-fitted thresholds lies between 0.16% and 0.96% in every cell, with no cell above 1% by point estimate; the worst cell is key 03 at 512 tokens, 0.96% [0.71, 1.27]. KGW cells range 0.24% to 0.96% and SynthID cells 0.16% to 0.66%. Empirical calibration at $n = 5,000$ per cell thus delivers the operating point the nominal threshold promised, provided the threshold is indexed by key and length.

H. What Calibration Costs in Detection

A per-key threshold that is far above the nominal one invites an obvious objection: for key 07 at 512 tokens it sits at $z > 6.40$ against a nominal 2.326, and a threshold that high might detect nothing. We can answer this from stored scores. A 1,000-output development screen generated watermarked text under the same model, the same watermark configuration and the same ten-key schedule, so the thresholds frozen on the null’s calibration split apply to it unchanged. Those thresholds were fitted on unwatermarked text alone, so using them on watermarked text is not circular. Detection rate here is the fraction of

watermarked outputs scoring above the threshold under their own key, on 50 outputs per cell.

Calibration does not disarm the detector (Figure 10a). The median detection rate across the ten keys falls from 99% to 97% at 512 tokens, from 98% to 95% at 256, and from 96% to 88% at 128. For key 07 at 512 tokens the false-positive rate falls from 48.4% to 0.54% while detection falls from 98% to 96%: an almost ninetyfold reduction in false positives for two points of detection. The worst cell in the whole set is key 08 at 128 tokens, where detection is 76% [62, 87].

The correction also runs in both directions (Figure 10b). Calibration raised the threshold in 27 of 30 cells and lowered it in three, and where it lowered the threshold, detection improved. Key 03 is the clearest case: its null sits well below the assumed one, its calibrated threshold at 512 tokens is $z > 1.49$ rather than 2.326, and its detection rate rises from 96% to 98%; at 128 tokens it rises from 78% to 86%. Averaged over cells, calibration costs 4.2 points of detection where it raises the threshold and gains 5.3 points where it lowers it. A conservative key is not harmless. It gives away detection power for a false-positive rate below the one it was asked for, and only a key-conditional threshold recovers that.

These are development-scale numbers on 50 outputs per cell, so the intervals are wide, and they are not a confirmatory detection claim. They are sufficient to answer the objection they were computed for: on this model and this watermark configuration, the price of a correctly calibrated false-positive rate is a few points of detection, not the loss of the watermark.

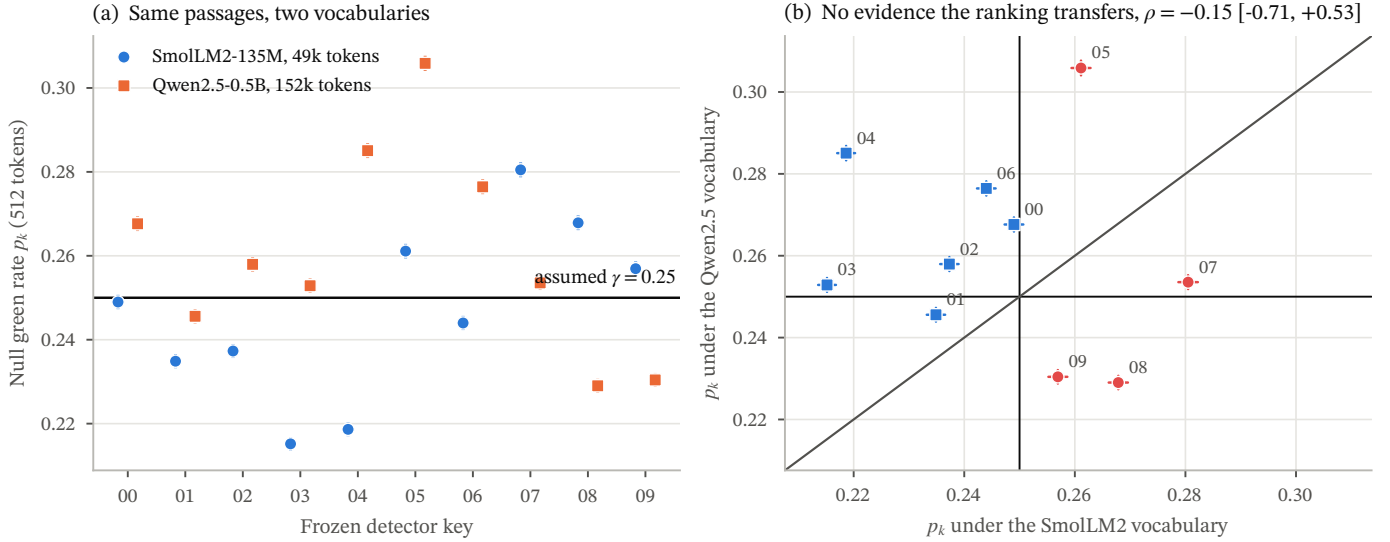
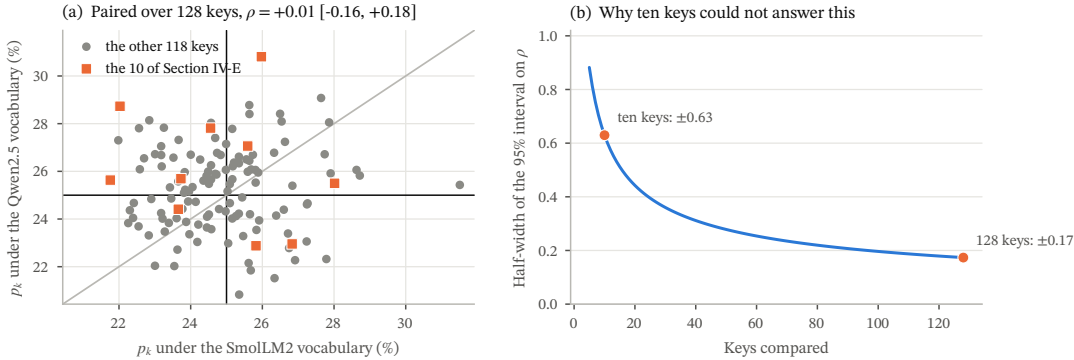


Fig. 7. The same 511 human-written Dolly passages at 512 tokens, scored with the same ten frozen KGW keys and the same watermark parameters under two vocabularies. (a) Null green rate p_k per key with exact 95% intervals; the rule marks the assumed $\gamma = 0.25$. The spread across keys does not narrow under the larger vocabulary. (b) p_k under the Qwen2.5 vocabulary against p_k under the SmolLM2 vocabulary, with exact intervals on both axes; the diagonal is equality and the rules mark γ . Colour is the same side-of- γ encoding as Figure 4a, taken from the SmolLM2 axis. Points fall off the diagonal in both directions; with ten keys the interval on ρ is wide, so this shows the absence of strong transfer rather than an established zero.

The key ranking across two vocabularies, powered



Identical human-written passages scored at 128 tokens under both vocabularies with the same 128 KGW keys and watermark parameters; passages are filtered on joint eligibility so the comparison is paired. Detector-only.

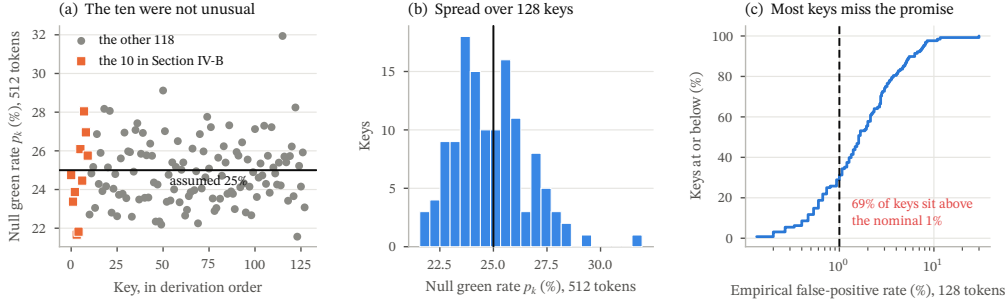
Fig. 8. The paired cross-vocabulary comparison of Figure 7, repeated over 128 keys at 128 tokens on the 800 passages both vocabularies support. (a) p_k under the Qwen2.5 vocabulary against p_k under the SmolLM2 vocabulary; the ten keys used throughout the paper are marked, and fall inside the wider population rather than beside it. The diagonal is equality. The cloud is round: $\rho = +0.01 [-0.16, +0.18]$. (b) Half-width of the 95% interval on ρ as a function of the number of keys compared, marking ten and 128. At ten keys the interval spans ± 0.63 and can exclude almost nothing, which is why Figure 7 supports only the absence of strong transfer; at 128 it spans ± 0.17 .

I. Fitting at the Acceptance Boundary Leaves No Familywise Margin

The same 60 cells were also subject to the prospective familywise rule of Section III-E, and that rule failed: 42 cells passed and 18 did not (14 KGW, 4 SynthID), even though every failed cell's point estimate lay between 0.58% and 0.96% and the largest Bonferroni-controlled upper bound was 1.48%. The reason is a design flaw rather than a held-out shift. Twenty-eight of 30 KGW calibration cells were fitted at exactly the maximum passing count of 28 exceedances, and KGW averaged 27.9 exceedances in calibration against 27.7 in confirmation (Figure 11a). At a

true rate equal to the boundary value $28/5,000 = 0.56\%$, a fresh 5,000-output cell has only about a 55% probability of staying at or below the passing count, so a rule that requires all 60 cells to pass was designed to fail. Prospective exact-binomial planning shows that more observations narrow the penalty but do not remove the need for a true operating rate below the target: under a conservative union bound, the largest true per-cell rate compatible with 95% whole-family pass probability rises from about 0.30% at 5,000 observations to 0.74% at 50,000 (Figure 11b). These are planning scenarios, not replacement thresholds. The failure is an instance of a known hazard rather than a local

Widening the key sample: the same schedule, 128 keys instead of ten



Detector-only scoring of human-written Dolly passages; keys derived by the schedule of Section III-B, of which conditions 0-9 are the ten used throughout the paper. (c) is drawn at 128 tokens, where every key is scored on all 1,500 passages.

Fig. 9. The same key schedule extended from ten conditions to 128, scored detector-only on 1,500 human-written Dolly passages. (a) Null green rate per key at 512 tokens, with the ten keys used elsewhere in the paper marked; they are not outliers. (b) The marginal distribution of p_k that ten keys could not show. (c) Empirical cumulative distribution of the false-positive rate at the nominal threshold, drawn at 128 tokens where every key is scored on all 1,500 passages; the dashed rule is the 1% the detector promises.

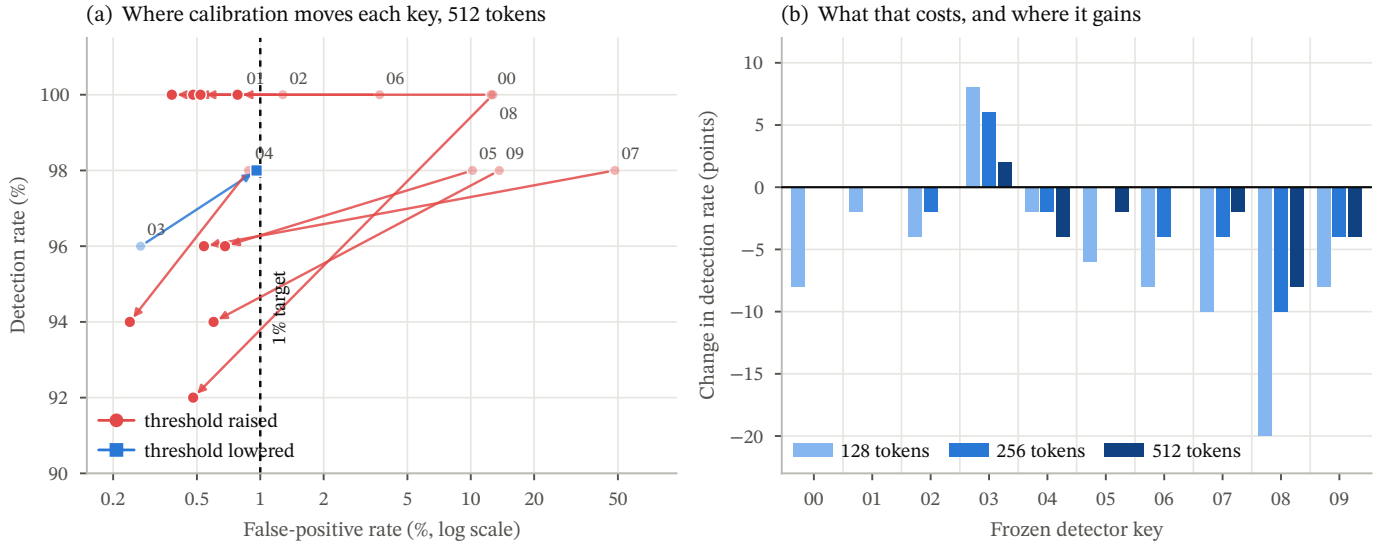


Fig. 10. (a) Operating point of each detector key at 512 tokens before and after per-key calibration, on a log false-positive axis. Arrows run from the nominal threshold to the calibrated one; labels sit at the nominal end. Colour marks whether calibration raised or lowered that key’s threshold. (b) Change in detection rate from calibration, by key and prefix length. The three cells where the threshold fell are the three where detection improved. Detection rates are development-scale, 50 watermarked outputs per cell.

accident: selecting a threshold on a calibration sample and then asserting a risk guarantee requires the selection itself to be treated as a multiple-testing problem, which is what the learn-then-test framework formalizes [3]. Our rule performed the selection and the guarantee at the same operating point, which is precisely the case that framework is built to rule out.

V. DISCUSSION

A. The Nominal Rate Is Not a Property of the Scheme

The i.i.d. Bernoulli(γ) null treats the green list as a random partition of the vocabulary that ordinary text does not prefer. It is not. The partition is uniform over vocabulary *types*, while text is drawn over *tokens*, and token frequency is famously far from uniform [30]. A quarter of the dictionary is therefore not a quarter of the words,

and how far it departs depends on which quarter the key selected. Each key’s partition lands differently on the frequent (context, token) pairs of English, so each key has its own p_k , and the z-test amplifies the difference by $\sqrt{T}$. Two consequences follow. A scheme-level false-positive rate is meaningless without the key: for the same model and threshold, one key gives 0.27% and another 48.4%. And longer texts do not help: for an inflated key the nominal test gets worse with length, because the bias grows with T while the assumed standard deviation does not.

That a population-level error rate can conceal large per-subject variation is not a new phenomenon, only a new instance of one. Speaker recognition named it three decades ago: aggregate error rates hid subjects whose individual rates were far worse, the “sheep, goats, lambs and wolves”

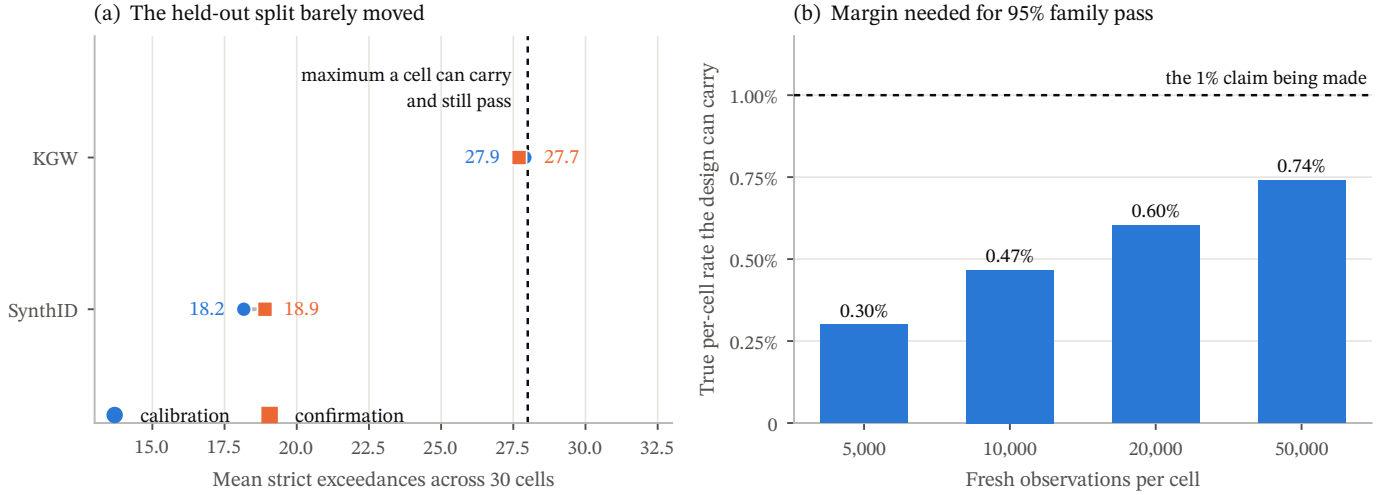


Fig. 11. (a) Mean strict exceedances across the 30 key-by-length cells of each scheme, on the calibration split and on the disjoint confirmation split, against the largest count a cell could carry and still pass. KGW’s pair sits on the boundary and barely moves, so the gate failure is not a held-out shift. (b) Prospective union-bound planning: the largest true per-cell rate compatible with at least 95% probability that all 60 cells pass, by fresh sample size. These are design scenarios, not replacement thresholds.

of the biometric menagerie [10], and face recognition later found error rates varying by orders of magnitude across demographic groups under a single reported threshold [14]. The discipline’s response in both cases was to stop reporting one aggregate number and to report the conditioning variable instead. For keyed watermark detection, the conditioning variable is the key.

B. The Bias Belongs to the Key and the Vocabulary

Human-written text under the same keys reproduces the ranking of p_k and the worst key, with smaller deviations from γ . This separates two explanations that the model-only data could not: the key effect is not an artefact of a tiny, repetitive model, and the model’s contribution is to concentrate its output on the pairs the key already favours. What the model does not decide is which key fails. Scoring those same passages under a second vocabulary does not narrow the spread of p_k , and over 128 keys the ranking does not survive it ($\rho = +0.01$ [-0.16, +0.18], Section IV-E), with the model, the keys and the text held fixed. The bias belongs to the key against the vocabulary it partitions; the model amplifies it but does not choose it. The Qwen2.5 replication, which varies model and vocabulary at once, is consistent with this and no longer needs a frequent-pair explanation of its own. All of these facts argue for the same practice. A deployment should measure p_k , or equivalently the null rate, for its own key and its own tokenizer on the text population it will score, including human text, rather than inherit a rate from the scheme. A rate measured for a key is not portable to a deployment that retokenizes, even when the key and the scheme are unchanged.

C. Calibration Works; Familywise Guarantees Need Margin

Per-key, per-length thresholds fitted once on 5,000 outputs delivered held-out rates below 1% in every cell. That

is the practical remedy, and it needs nothing beyond an unwatermarked corpus from the relevant population and a key schedule. It is also cheap in the currency practitioners care about: a few points of detection where the threshold rises, and more detection than the nominal threshold gave where it falls. The prospective familywise rule we also imposed failed, and the failure is instructive. Selecting the least conservative threshold that satisfies a calibration bound places every cell at the acceptance boundary, where a fresh sample passes with probability near one half; requiring all 60 to pass then has little chance of success however valid the multiplicity control. Protocols that want a simultaneous guarantee should declare both the final error target and a lower design rate that gives the family adequate pass probability, and should size the calibration corpus to that design rate rather than to the target.

D. What the Remedy Costs, and What a Cheaper One Would Need

Per-key, per-length calibration invites the objection that it is expensive: a fresh unwatermarked corpus for every key, length and tokenizer. The measurements here make it cheaper than that. What the threshold needs is p_k , and p_k is a property of the key against a text population, not of any generation procedure. Section IV-D estimates it on human-written passages with nothing generated and no model weights loaded, and Section IV-C shows that substituting the measured p_k for γ predicts the mean null z of all 30 cells at $r = 0.9999$. A deployment can therefore calibrate against any corpus that resembles what it will score, including text it already holds, rather than sampling thousands of generations per key.

A purely analytical correction would be cheaper still, and is the natural next question: if the bias comes from the key’s partition meeting the frequency structure of ordinary

text, predict p_k from that structure and skip the corpus pass. We do not offer one, and the variant measured here is the reason to be careful. Under SelfHash the green list at each position depends on the candidate token as well as the preceding context, so p_k is not a marginal of the unigram distribution over a fixed partition and cannot be recovered by reweighting one. A context-only variant such as LeftHash is a more tractable starting point, where the partition is fixed once the preceding token is known and p_k becomes an expectation over the bigram distribution. Whether that estimate is accurate enough to replace a measured threshold is an open question this paper does not settle; the released score files and the per-position trace exporter are the material for testing it.

E. Implications for Operational Use

A keyed detector score is evidence of compatibility with a specified embedding and detection process. It is not proof of authorship, misconduct, truth, or intent. Operational claims should name the scheme variant, the key policy, the length range, the null population on which the threshold was calibrated, the uncertainty procedure, and the consequences of a false positive. On the evidence here, a claim that rests on the nominal rate alone is unsupported for KGW-style detectors.

VI. LIMITATIONS

The primary experiments use small language models and pinned English prompt sources; the size of the effect will differ for other models, languages, and domains, although the mechanism does not depend on model size. Findings apply to the explicitly named variants and parameter settings rather than to every implementation called KGW or SynthID. The Qwen2.5 replication is small ($n = 104$) and establishes only that the failing keys differ. The vocabulary comparison of Section IV-E uses two specific tokenizers, so it shows that the vocabulary matters but cannot attribute that to vocabulary size as such rather than to the particular merges each tokenizer learned; its ten-key cell at 512 tokens rests on the 511 passages both vocabularies support. The paired sweep that widens that comparison to 128 keys scores 800 jointly eligible passages, so its own 512-token cell rests on 93; the rank correlation there is the least well-supported of the three lengths, and the 128-token cell, at 800 passages, is the one to quote. The human-text null has 554 passages at 512 tokens, and its longer passages are largely reference text rather than answers. SynthID-Text was evaluated against a naive-independence reference because its mean-g detector has no nominal z ; its calibrated behaviour is in Appendix A. The detection rates in Section IV-H come from a 50-output-per-cell development screen, so they price the remedy at development scale rather than confirming a detection claim. No robustness, attack or paraphrase conclusion is available.

VII. ETHICS AND RESPONSIBLE USE

False accusations based on automated text detection can cause material harm, and the harm is not distributed evenly: detectors built for this purpose have been shown to misclassify non-native English writing at far higher rates than native writing [23]. A keyed watermark detector is a different mechanism, but the failure documented here has the same shape, an error rate that is quoted as a single number and is in fact conditional on something the reader is not told. Accordingly, all claims in this paper remain conditional on named implementations, keys, lengths, populations, and decision rules. We discourage use of watermark detection as standalone evidence in disciplinary, employment, legal, or authorship decisions. Raw generated continuations will be reviewed for memorized personal, copyrighted, or unsafe content before any public release.

VIII. REPRODUCIBILITY AND ARTIFACT AVAILABILITY

The repository records frozen configurations, model and dataset revisions, prompt selection manifests, key schedules, implementation fingerprints, per-position audit formats, the frozen calibration thresholds, and the confirmation record. Every table and figure in this paper is generated deterministically from stored scores by a fixed script; no analysis required new generation. Model weights and cached datasets are not redistributed; deterministic retrieval instructions and immutable revisions are supplied instead.

The release is at <https://github.com/nisaharan/ai-watermarks-3d> and includes source code, tests, aggregate artifacts, prompt row identifiers and hashes, tables, figures, and parity fixtures. It also carries the three compressed score files the analyses read and the aggregated cells of both key sweeps, so a clone regenerates every table and figure in this paper without generating any text; this was checked by cloning the release and rebuilding the paper from it alone. The text-level tables carry token statistics and a SHA-256 per text, never the text itself. Raw Dolly text requires CC BY-SA 3.0 attribution and share-alike treatment. Raw generated continuations and development runs that support no claim in this paper are retained privately for audit and excluded from the reported package.

IX. CONCLUSION

For canonical KGW SelfHash, the nominal 1% false-positive threshold is neither 1% nor a constant: it is a key- and length-dependent quantity that ranged from 0.27% to 48.4% on 10,000 unwatermarked outputs, because each key induces its own null green rate on real text and the detector's z -test scales the resulting bias by $\sqrt{T}$. Human-written text reproduces the key ranking, so the effect is a property of the key rather than the model, and rescoring the same passages under a second vocabulary leaves no evidence that the ranking survives the change, so the quantity that matters is the pair of key and tokenizer. Widening the key

sample from ten to 128 shows this is not a property of an unlucky draw: 69% of keys sit above the nominal 1% at 128 tokens and 53% can be shown to from 1,500 passages alone, with the ten keys used here sitting unremarkably inside that distribution. Empirical calibration indexed by key and length removes it, and it needs a corpus rather than a generation campaign, because what the threshold depends on is p_k . Evaluations and deployments of green-list watermarks should therefore report the key schedule, the tokenizer and the calibration population as part of the experimental condition, and should treat a nominal rate as a hypothesis to be tested rather than a property to be quoted.

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APPENDIX A

CALIBRATED PER-CELL HELD-OUT FALSE-POSITIVE RATES

Table III lists the 30 canonical KGW cells and Table IV the 30 SynthID-Text cells. Each row gives the threshold fitted on the calibration split, the strict exceedance count on both splits, and the held-out rate on the confirmation split with its exact interval. No cell’s held-out point estimate exceeds 1%.

APPENDIX B

DATA SOURCES AND EVIDENTIAL ROLE

Table V lists every corpus the paper uses and what each one is for. All scoring is detector-only on stored token ids.

APPENDIX C

INTERPRETATION CHECKLIST

- Every rate is conditional on the named variant, key, prefix length, threshold, and null population; none transfers to another key by default.
- A cell that failed the familywise rule in Section IV-I is not evidence that its true rate exceeds 1%; its point estimate is in Appendix A.
- The SynthID-Text figures are a naive-independence reference, not a nominal claim by that scheme.
- Canonical and Transformers variant scores are never pooled.
- The detection rates are a development screen of 50 outputs per cell, reported to price calibration; they are not a confirmatory detection claim.
- Robustness to editing, paraphrase or attack was not evaluated.

TABLE III

CANONICAL KGW SELFHASH (THRESHOLDS INDEXED BY KEY AND LENGTH). HELD-OUT FALSE-POSITIVE RATE OF THE ONCE-FITTED THRESHOLD ON THE 5,000-OUTPUT CONFIRMATION SPLIT, WITH EXACT TWO-SIDED 95% CLOPPER-PEARSON INTERVALS. CALIB. AND CONF. ARE STRICT EXCEEDANCE COUNTS ON THE CALIBRATION AND CONFIRMATION SPLITS; THE BONFERRONI-CONTROLLED ONE-SIDED EXACT UPPER BOUND HAD TO BE AT MOST 1% (28 EXCEEDANCES) FOR THE CELL TO PASS.

Key	T	Threshold	Calib.	Conf.	Held-out FPR (%) [95% CI]	Upper bound (%)	Pass
kgw-00	128	3.8636	28	27	0.54 [0.36, 0.78]	0.95	yes
kgw-00	256	4.3585	28	27	0.54 [0.36, 0.78]	0.95	yes
kgw-00	512	5.0208	28	26	0.52 [0.34, 0.76]	0.92	yes
kgw-01	128	2.4271	27	28	0.56 [0.37, 0.81]	0.98	yes
kgw-01	256	2.6469	28	17	0.34 [0.20, 0.54]	0.69	yes
kgw-01	512	2.7107	28	19	0.38 [0.23, 0.59]	0.74	yes
kgw-02	128	2.7406	28	14	0.28 [0.15, 0.47]	0.60	yes
kgw-02	256	3.0645	28	17	0.34 [0.20, 0.54]	0.69	yes
kgw-02	512	3.0632	28	24	0.48 [0.31, 0.71]	0.87	yes
kgw-03	128	1.9261	26	32	0.64 [0.44, 0.90]	1.08	no
kgw-03	256	1.7457	28	34	0.68 [0.47, 0.95]	1.13	no
kgw-03	512	1.4927	28	48	0.96 [0.71, 1.27]	1.48	no
kgw-04	128	2.4444	28	24	0.48 [0.31, 0.71]	0.87	yes
kgw-04	256	2.5404	28	35	0.70 [0.49, 0.97]	1.15	no
kgw-04	512	3.2515	28	12	0.24 [0.12, 0.42]	0.55	yes
kgw-05	128	3.6977	28	34	0.68 [0.47, 0.95]	1.13	no
kgw-05	256	4.1435	28	34	0.68 [0.47, 0.95]	1.13	no
kgw-05	512	4.8658	28	34	0.68 [0.47, 0.95]	1.13	no
kgw-06	128	2.9673	28	32	0.64 [0.44, 0.90]	1.08	no
kgw-06	256	3.0995	28	37	0.74 [0.52, 1.02]	1.20	no
kgw-06	512	3.3158	28	39	0.78 [0.56, 1.06]	1.25	no
kgw-07	128	4.8488	28	23	0.46 [0.29, 0.69]	0.85	yes
kgw-07	256	5.6614	28	16	0.32 [0.18, 0.52]	0.66	yes
kgw-07	512	6.3967	28	27	0.54 [0.36, 0.78]	0.95	yes
kgw-08	128	4.3146	28	27	0.54 [0.36, 0.78]	0.95	yes
kgw-08	256	4.9333	28	30	0.60 [0.41, 0.86]	1.03	no
kgw-08	512	5.6159	28	24	0.48 [0.31, 0.71]	0.87	yes
kgw-09	128	3.7778	28	29	0.58 [0.39, 0.83]	1.00	no
kgw-09	256	4.2699	28	31	0.62 [0.42, 0.88]	1.05	no
kgw-09	512	4.7962	28	30	0.60 [0.41, 0.86]	1.03	no

TABLE IV

CANONICAL SYNTHID-TEXT (ONE SHARED THRESHOLD PER LENGTH). HELD-OUT FALSE-POSITIVE RATE OF THE ONCE-FITTED THRESHOLD ON THE 5,000-OUTPUT CONFIRMATION SPLIT, WITH EXACT TWO-SIDED 95% CLOPPER–PEARSON INTERVALS. CALIB. AND CONF. ARE STRICT EXCEEDANCE COUNTS ON THE CALIBRATION AND CONFIRMATION SPLITS; THE BONFERRONI-CONTROLLED ONE-SIDED EXACT UPPER BOUND HAD TO BE AT MOST 1% (28 EXCEEDANCES) FOR THE CELL TO PASS.

Key	T	Threshold	Calib.	Conf.	Held-out FPR (%) [95% CI]	Upper bound (%)	Pass
synthid-00	128	0.5456	12	12	0.24 [0.12, 0.42]	0.55	yes
synthid-00	256	0.5344	27	21	0.42 [0.26, 0.64]	0.79	yes
synthid-00	512	0.5293	23	16	0.32 [0.18, 0.52]	0.66	yes
synthid-01	128	0.5456	10	14	0.28 [0.15, 0.47]	0.60	yes
synthid-01	256	0.5344	15	20	0.40 [0.24, 0.62]	0.77	yes
synthid-01	512	0.5293	16	21	0.42 [0.26, 0.64]	0.79	yes
synthid-02	128	0.5456	9	15	0.30 [0.17, 0.49]	0.63	yes
synthid-02	256	0.5344	16	14	0.28 [0.15, 0.47]	0.60	yes
synthid-02	512	0.5293	24	16	0.32 [0.18, 0.52]	0.66	yes
synthid-03	128	0.5456	28	30	0.60 [0.41, 0.86]	1.03	no
synthid-03	256	0.5344	20	32	0.64 [0.44, 0.90]	1.08	no
synthid-03	512	0.5293	28	31	0.62 [0.42, 0.88]	1.05	no
synthid-04	128	0.5456	19	17	0.34 [0.20, 0.54]	0.69	yes
synthid-04	256	0.5344	28	22	0.44 [0.28, 0.67]	0.82	yes
synthid-04	512	0.5293	27	25	0.50 [0.32, 0.74]	0.90	yes
synthid-05	128	0.5456	15	9	0.18 [0.08, 0.34]	0.46	yes
synthid-05	256	0.5344	14	20	0.40 [0.24, 0.62]	0.77	yes
synthid-05	512	0.5293	17	22	0.44 [0.28, 0.67]	0.82	yes
synthid-06	128	0.5456	6	8	0.16 [0.07, 0.32]	0.43	yes
synthid-06	256	0.5344	14	14	0.28 [0.15, 0.47]	0.60	yes
synthid-06	512	0.5293	19	16	0.32 [0.18, 0.52]	0.66	yes
synthid-07	128	0.5456	17	12	0.24 [0.12, 0.42]	0.55	yes
synthid-07	256	0.5344	17	17	0.34 [0.20, 0.54]	0.69	yes
synthid-07	512	0.5293	23	21	0.42 [0.26, 0.64]	0.79	yes
synthid-08	128	0.5456	11	13	0.26 [0.14, 0.44]	0.57	yes
synthid-08	256	0.5344	27	25	0.50 [0.32, 0.74]	0.90	yes
synthid-08	512	0.5293	18	33	0.66 [0.45, 0.93]	1.10	no
synthid-09	128	0.5456	13	12	0.24 [0.12, 0.42]	0.55	yes
synthid-09	256	0.5344	16	19	0.38 [0.23, 0.59]	0.74	yes
synthid-09	512	0.5293	16	20	0.40 [0.24, 0.62]	0.77	yes

TABLE V

EVERY CORPUS USED IN THIS PAPER AND THE ROLE IT PLAYS. ALL SCORING IS DETECTOR-ONLY ON STORED TOKEN IDS.

Corpus	Size	Role
SmolLM2-135M unwatermarked outputs	10,000	Primary null (Sections IV-B–IV-C); also the calibration and confirmation splits of 5,000 each (Sections IV-G–IV-I).
SmolLM2 development pilot	500	Secondary run; reproduces the primary key profile.
Qwen2.5-0.5B replication	104	Secondary run; shows that the failing keys differ once model and tokenizer both change.
Human-written Dolly passages	5,000	Human-text null (Section IV-D) and the second-vocabulary comparison (Section IV-E); detector-only, no generation.
Human-written Dolly passages, key sweep	1,500	128 keys from the same schedule (Section IV-F); detector-only, KGW only.
Human-written Dolly passages, paired vocabulary sweep	800	The same 128 keys under both tokenizers, on passages both vocabularies support (Section IV-E); detector-only, KGW only.
Watermarked positive screen	1,000	Detection rates at the calibrated thresholds (Section IV-H); development scale, 50 outputs per cell.
Development bias brackets	1,400	Out of scope; support no claim here.